WEEK 07 / GAMING'S IMPACT ON PEOPLE AND SOCIETY

Gaming and Medicine

The use of gaming in medical therapy and mental health.

This week we explore the use of gaming in medical therapy and mental health, discussing how video games can be utilized for therapeutic purposes and their potential benefits for mental well-being.

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31 RESOURCES / 1 TOPIC

TOPIC 01

Games, Health, and Medicine

REQUIRED

The Expanding Role of Video Games in Medicine

ARTICLE | ~5 MIN

https://opmed.doximity.com/articles/the-expanding-role-of-video-games-in-medicine

Written by medical student Star Chen, this essay explores the increasing applications of video games in the medical field, serving both as therapeutic interventions and training mechanisms. While acknowledging the risks of addiction, the author highlights the significant cognitive, emotional, and social benefits of casual gaming, such as improved focus, patience, and problem-solving. The article details how "serious games" and extended reality (augmented and virtual reality) are being used to enhance surgical skills, improve spatial awareness, and treat various conditions, particularly neurological and psychological ones like Parkinson's disease, cerebral palsy, and pain management. Despite limitations like cost and limited content, the author expresses optimism for the future expansion of video games in medicine.

- A meta-analysis found that playing first-person shooter games can improve players' spatial awareness and attention allocation at a rate comparable to or even faster than university-level lectures designed to teach these skills, with better long-term retention.
- A history of gaming has been associated with improved metrics in surgery and laparoscopy, suggesting video games could serve as "adjunctive training for surgical skills".
- Video games are effective in treating balance and mobility issues in patients with Parkinson's disease and cerebral palsy by keeping them engaged and enthusiastic, while providing visual and auditory feedback on performance.
- The video game industry was worth $90.1 billion in 2023, triple its value a decade prior, and is projected to continue growing.

LO 3: Students will explore the applications of video games beyond entertainment, including their use in medicine, education, and social engagement.

REQUIRED

When A Video Game Becomes An Addiction

TED TALK | ~15 MIN

https://www.ted.com/talks/khurram_sadiq_when_a_video_game_becomes_an_addiction?subtitle=en

In this TEDx presentation, Dr. Khurram Sadiq shares his personal struggle with compulsive gaming during his medical training to highlight a growing public health concern. The presentation details how medical professionals are increasingly recognizing the severe physical and psychological consequences tied to contemporary interactive entertainment, particularly focusing on issues like physical blood clots, extreme emotional outbursts, and insular online communities. Ultimately, the speaker aims to raise awareness about Internet Gaming Disorder and urges players to maintain a healthy equilibrium between virtual worlds and everyday life.

LO 4: Students will be able to discuss ethical concerns surrounding gaming, including issues related to violence, addiction, monetization practices, labor concerns, and virtual identities.

REQUIRED

Video Games Offer the Potential of “Experiential Medicine”

ARTICLE/YOUTUBE | ~8 MIN

https://www.ucsf.edu/news/2022/10/423941/video-games-offer-potential-experiential-medicine

This article from UC San Francisco's Neuroscape Center discusses their pioneering work in developing video game interventions that can improve cognitive function in aging adults. Co-creator Adam Gazzaley describes this as a new form of "experiential medicine" where games use adaptive closed-loop algorithms to personalize difficulty, thereby enhancing crucial cognitive processes like short-term memory, attention, and long-term memory. These interventions, which have shown benefits in randomized clinical trials, are being adapted for clinical populations and have potential applications for conditions such as ADHD, Parkinson's, and multiple sclerosis.

- Scientists at UC San Francisco's Neuroscape Center have developed video game interventions that use adaptive closed-loop algorithms to automatically adjust difficulty based on a player's performance, preventing overwhelm for less skilled players while still challenging others.
- Their musical rhythm game, developed with drummer Mickey Hart, not only taught 60-79-year-old participants how to drum but also improved their ability to remember faces, showing increased brain activity in regions linked to visual memory.
- The "Body Brain Trainer" game, which improved blood pressure, balance, and attention in older adults, incorporated a heart rate monitor feedback system to ensure participants were exercising at the correct intensity.
- These digital training methods, including games for multitasking, working memory, and spatial navigation, have demonstrated their ability to enhance waning cognitive faculties in older adults through randomized clinical trials.

LO 3: Students will explore the applications of video games beyond entertainment, including their use in medicine, education, and social engagement.

REQUIRED

How Health Systems are Using Video Games for Continuing Medical Education

ARTICLE | ~10 MIN

https://www.healthleadersmedia.com/innovation/how-health-systems-are-using-video-games-continuing-medical-education

This article from HealthLeaders Media features an interview with Dr. Eric Gantwerker, a pediatric otolaryngologist and medical director for Level Ex, a video game studio specializing in continuing medical

education (CME). Dr. Gantwerker explains that video games are increasingly used in healthcare not just for patient education, but also as powerful educational tools for clinicians. He highlights how games are built on deep understanding of brain stimulation and learning, creating intrinsically motivated environments that can improve knowledge retention and skill development more effectively than traditional passive learning methods. Challenges include the public perception of games as child-like entertainment and the need for expert game designers.
- Video games are inherently designed with a deep understanding of how to activate and stimulate the brain to induce learning, drawing on sociology, psychology, and cognitive sciences.
- Unlike traditional medical learning and CME, which are often extrinsically motivated requirements, well-designed games create an intrinsically motivated, entertaining environment that healthcare professionals actively want to engage with, leading to better learning.
- As of August 2022, there are 2.6 billion video game players globally, with an average age of 36, yet a significant perception challenge persists that games are only for children and are "brain-numbing".
- Studies have shown that playing action video games at a young age is associated with better visuospatial skills and improved performance on laparoscopic surgery-based tasks.

LO 3: Students will explore the applications of video games beyond entertainment, including their use in medicine, education, and social engagement.

REQUIRED

Zach Wigal on helping hospitals through gaming

ARTICLE | ~18 MIN

https://www.michigandaily.com/arts/digital-culture/zach-wigal-on-helping-hospitals-through-gaming/

This interview provides insights into Gamers Outreach, a charity founded by Zach Wigal that aims to improve the quality of life for children in hospitals by providing access to video games. Wigal recounts the organization's origin from a Halo tournament being shut down by police due to negative perceptions of gaming, leading him to demonstrate gaming's positive community aspects. He details their core initiatives: the "Gamers for Giving" LAN party, which raises significant funds, and their "GO Karts" (portable gaming kiosks) and "Player 2" program (volunteers assisting hospital staff with gaming equipment). Wigal emphasizes how mainstream video games have become and their crucial role in providing socialization, entertainment, creativity, and a sense of agency for hospitalized patients.
- Gamers Outreach was founded in 2007 after Zach Wigal's Halo tournament was shut down by a police officer who believed video games were "corrupting the lives of America's youth".
- Their annual "Gamers for Giving" LAN party gathered hundreds of gamers and raised over $1.25 million in April 2024 to support children's hospitals worldwide.
- "GO Karts" are portable video game kiosks equipped with consoles, monitors, and games, which are estimated to benefit over five million kids per year across all deployed units.
- A 13-year-old boy awaiting a heart transplant for nine months at Children's Hospital Los Angeles used Minecraft to interact with friends and maintain a "digital playground," which his mom credited with helping him "keep it together" during his long hospital stay.

LO 2: Students will be able to critically assess how video games reflect and shape cultural values, norms, and societal issues.
LO 3: Students will explore the applications of video games beyond entertainment, including their use in medicine, education, and social engagement.
LO 6: Students will understand the variety of gaming experiences, from indie games to eSports, and will recognize the contributions of diverse communities within the gaming industry.

REQUIRED

The Benefits of Playing Video Games

ARTICLE | ~40 MIN

https://maine.primo.exlibrisgroup.com/permalink/01MAINE_INST/1oa9cn5/cdi_proquest_miscellaneous_1636462259

This academic paper advocates for a more balanced psychological perspective on video games, moving beyond the prevalent focus on negative impacts like violence and addiction to explore their substantial benefits. It synthesizes research on four key domains: cognitive (attention, spatial skills, problem-solving, creativity), motivational (persistence, optimism, resilience to failure), emotional (mood management, positive emotions like "fiero" and "flow"), and social (prosocial behavior, cooperation, civic engagement). The authors propose mechanisms for how these gaming experiences translate to real-world psychosocial benefits and call for new research programs, especially in mental health interventions, recognizing games' immersive and engaging nature.
- In the United States, 91% of children between the ages of 2 and 17 play video games, with up to 99% of boys and 94% of girls playing these games (as of 2011 studies).
- Playing shooter video games has been shown to result in faster and more accurate attention allocation, higher spatial resolution in visual processing, and enhanced mental rotation abilities, with spatial skill improvements comparable to formal high school and university courses.
- Video games are proposed as an ideal training ground for acquiring an "incremental theory of intelligence" (the belief that intelligence is malleable and can be improved through effort), as they provide concrete, immediate feedback that rewards continual effort.
- "Fiero" (intense pride after succeeding against great adversity) and "flow" (immersion in an intrinsically rewarding activity with a high sense of control) are intense positive emotional experiences that gamers often report seeking and experiencing.
- The game Re-Mission, designed for child cancer patients, significantly improved adherence to treatment protocol, self-efficacy, and cancer-related knowledge in a randomized controlled study across 34 medical centers.

LO 3: Students will explore the applications of video games beyond entertainment, including their use in medicine, education, and social engagement.

REQUIRED

Can a video game save a life?

TED TALK | ~14 MIN

https://www.youtube.com/watch?v=6rLJaVohSRQ&ab_channel=TEDxTalks

This TEDx talk by Johnny Chiodini, a Senior Video Producer for Eurogamer.net, explores the profound ways video games intersect with and positively impact mental health. Chiodini shares his personal journey with depression and anxiety, emphasizing the importance of open dialogue about mental health, particularly for men, due to prevailing stigma. He asserts that video games can "save a life" by providing distraction and a sense of accomplishment, fostering supportive communities, and serving as direct therapeutic tools. He advocates for the gaming industry and individuals to engage more openly with mental health topics.
- Johnny Chiodini has suffered from depression and anxiety since he was about 14 years old, but was not diagnosed until 2012, finding cognitive behavioral therapy and antidepressants "extremely useful".
- In 2014, League of Legends was played by 67 million people worldwide every month, with 7.5 million people playing simultaneously during peak hours-only 1 million less than the population of London at the time.
- The concept of a "sad game" suggests that people suffering from depression sometimes gravitate toward a specific video game for distraction when things are particularly bad, offering a "breather" from their immediate problems.
- Examples of games directly addressing mental health include Life is Strange (themes of depression and suicide), Papo and Yo (addiction's impact on family), and Flowy (designed for anxiety and panic attacks to encourage regular breathing and calming). Virtual Iraq is a virtual therapeutic experience for combat veterans

suffering from PTSD, using virtual reality and even piped smells to expose patients to traumatic events in a controlled environment.

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REQUIRED

How Years Of Gaming Affects Your Brain

YOUTUBE | ~25 MIN

https://www.youtube.com/watch?v=xaICKlp9kQc&ab_channel=HealthyGamerGG

The video explores the neurological effects of prolonged video game engagement, primarily focusing on the imbalance between dopamine and serotonin. It suggests that excessive gaming leads to a dopaminergic dependence and serotonergic aversion, resulting in gamers feeling stuck, lacking confidence, and struggling with real-world interactions. Furthermore, the video explains how gaming suppresses negative emotional processing, hindering learning and problem-solving abilities by atrophying the brain's capacity for operationalization. Ultimately, it proposes strategies like sitting with negative emotions and breaking down complex tasks to reverse these neurological deficits and foster personal growth.

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EXPLORE FURTHER

SuperBetter

BOOK

https://maine.primo.exlibrisgroup.com/permalink/01MAINE_INST/1i25tbk/alma991004860959708681

Book description: “In this book, McGonigal reveals a decade’s worth of scientific research into the ways all games-including videogames, sports, and puzzles-change how we respond to stress, challenge, and pain. She explains how we can cultivate new powers of recovery and resilience in everyday life simply by adopting a more “gameful” mind-set.”

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EXPLORE FURTHER

Getting gamers : the psychology of video games and their impact on the people who play them

BOOK

https://maine.primo.exlibrisgroup.com/permalink/01MAINE_INST/1i25tbk/alma991005765099708681

Book description: “This book will show that rather than being a waste of time, video games can help us develop skills, make friends, succeed at work, form good habits, and be happy. Taking the time to learn what's happening in our heads as we play and shop allows us to approach games and gaming communities on our own terms and get more out of them.”

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EXPLORE FURTHER

Mind at play : the psychology of video games

BOOK

https://maine.primo.exlibrisgroup.com/permalink/01MAINE_INST/1i25tbk/alma991034969089708681

This 1983 book… “Analyzes the fascination of computer games, discussing reinforcement, the arcade subculture, et cetera”

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EXPLORE FURTHER

The Psychology of Video Games

BOOK

https://www.amazon.com/Psychology-Video-Games-Everything/dp/0367493136

Book description: “The Psychology of Video Games introduces the curious reader to the relationship between psychology and video games from the perspective of both game makers and players.”

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EXPLORE FURTHER

Foldit

GAME

https://fold.it/

“Foldit is a revolutionary crowdsourcing computer game enabling you to contribute to scientific research. Learn the science behind Foldit and how your playing can help.“
Read articles on Foldit and other medical simulation games or check out the game itself.

LO 3: Students will explore the applications of video games beyond entertainment, including their use in medicine, education, and social engagement.

EXPLORE FURTHER

NOVA scienceNOW profile of FoldIt: A Protein Puzzle Game from PBS

ARTICLE

https://mainepublic.pbslearningmedia.org/resource/nvsn6.sci.tech.foldit/foldit-a-protein-puzzle-game/

"In this video excerpt from NOVA scienceNOW, learn about Foldit, an online game in which players contribute to scientific research by solving puzzles. The objective of the game is to fold proteins into optimal 3D configurations. By identifying the structural configurations of proteins, scientists can make advances in the treatment of diseases and other problems. One of Foldit’s creators describes the challenge of creating a game that nonscientists could play and take an interest in without necessarily understanding the science of proteins. In 2011, Foldit players identified the structure of a protein that helps HIV reproduce. This was a major accomplishment and will help researchers develop treatments for HIV."

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EXPLORE FURTHER

Foldit: Competitive Protein Folding for Medical Science

ARTICLE

https://www.engr.washington.edu/facresearch/highlights/cse_foldit.html

Focusing on the 2008 launch of the game, this piece details the initial vision and technical mechanics behind turning protein structure prediction into a competitive crowdsourced sport. Unlike earlier distributed computing projects like Rosetta@home that only harvested idle computer processing cycles, Foldit actively capitalizes on human 3D spatial intuition. The article highlights unique design constraints: in-game scores directly correspond to the physical potential energy required to hold a protein's shape rather than arbitrary gaming points, preventing developers from offering artificial hints. It also outlines early aspirations to discover non-biologist "spatial prodigies" and apply human-designed synthetic proteins toward environmental challenges like absorbing carbon dioxide and breaking down toxic waste.

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EXPLORE FURTHER

Gamers succeed where scientists fail

ARTICLE

https://www.washington.edu/news/2011/09/19/gamers-succeed-where-scientists-fail/

This article highlights a specific scientific breakthrough where Foldit players solved the 3D crystal structure of the M-PMV retroviral protease-an AIDS-like monkey virus enzyme that had stumped biochemists for over a decade-in just three weeks. By identifying key molecular surfaces on the enzyme that serve as targets for retroviral drugs, the players earned co-authorship alongside researchers in Nature Structural & Molecular Biology. A key unique insight emphasized is the gamers' first-time use of Foldit's "Alignment Tool," which allowed players to copy and combine regions from multiple known molecules into a single workable hybrid structure.

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EXPLORE FURTHER

Behind the Scenes of Foldit, Pioneering Science Gamification

ARTICLE

https://www.americanscientist.org/blog/from-the-staff/behind-the-scenes-of-foldit-pioneering-science-gamification

This 2012 retrospective interview with lead designer Seth Cooper provides behind-the-scenes insights into Foldit's design evolution, describing it as a blend of "3D Tetris" and "Minecraft" that functions as a continuous live experiment using telemetry and weekly updates. It details player-driven innovations, such as user-scripted automated recipes in "The Cookbook" (like the popular "Blue Fuse" script), which organically replicated an algorithm developed independently in a biochemistry lab. Furthermore, it showcases advanced de novo protein engineering, including a player-designed Diels-Alder reaction catalyst enzyme that increased efficiency twentyfold through a drastic 13-amino-acid insertion that biochemists would never have attempted. Cooper also discusses adapting Foldit for classroom curricula, experimenting with Kinect motion controls, and expanding the concept to self-assembling DNA nanomachines.

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EXPLORE FURTHER

Best Medical Games, Ranked

ARTICLE

https://www.thegamer.com/best-medical-video-games-ranked/

This article explores the diverse landscape of medical video games, highlighting how these titles shift the focus from combat to healing and healthcare management. It categorizes various games based on their gameplay mechanics, ranging from realistic surgical simulations and educational tools to humorous management sims and psychological horror. Some featured titles emphasize professional training and anatomy, while others offer fantastical scenarios involving shrinking surgeons or janitors cleaning up futuristic laboratories. Beyond simple entertainment, the article notes that these games appeal to both medical professionals and casual players by tapping into curiosity about human biology. Ultimately, the collection demonstrates how the genre uses interactive technology to explore the complexities of medicine through a variety of unique and engaging lenses.

LO 3: Students will explore the applications of video games beyond entertainment, including their use in medicine, education, and social engagement.

EXPLORE FURTHER

A video game that helps us understand loneliness

TED TALK | ~13 MIN

https://www.youtube.com/watch?v=oMBVxhO57JE

In this TED talk, artist Cornelia Geppert discusses how she channeled her personal experience with loneliness into creating an interactive story in the form of a video game called "Sea of Solitude". The game features a protagonist named Kay, who is suffering from loneliness so intensely that her inner feelings manifest externally, turning her into a monster. The game's world represents Kay's mind, flooded by her tears and with changing weather reflecting her mood. Geppert explains that the game explores various manifestations of loneliness, including social exclusion, loneliness within relationships, and loneliness due to mental health issues like depression. The game's monsters are personifications of Kay's emotions, such as Self-Doubt and Self-Destruction, which players learn to understand and balance rather than simply eliminate. Geppert emphasizes that the game is not a substitute for therapy but an artistic expression that helped many players feel less alone and even seek therapy themselves.
- The game "Sea of Solitude" portrays the protagonist, Kay, transforming into a monster due to her strong feelings of anger, hopelessness, and worthlessness, which is a representation of the artist's own struggles with loneliness.
- The game's environment is designed to reflect Kay's emotional state, with the world being flooded by her tears and the weather changing according to her mood.
- After the game's release, thousands of fans shared their stories, stating they felt less alone and many even sought therapy after playing "Sea of Solitude".

LO 3: Students will explore the applications of video games beyond entertainment, including their use in medicine, education, and social engagement.

EXPLORE FURTHER

Effects of Interactive Physical-Activity Video-Game Training on Physical and Cognitive Function in Older Adults

ARTICLE | ~36 MIN

https://maine.primo.exlibrisgroup.com/permalink/01MAINE_INST/1oa9cn5/cdi_hal_primary_oai_HAL_hal_00736985v1

This journal article details a study assessing the potential of exergames (physically simulated sport video games) to improve physical and cognitive function in older adults. The study hypothesized that exergame training would lead to improved performance in the games themselves and transfer to a wide range of

neuropsychological tests, including executive control, visuospatial, and processing speed tasks. Participants, primarily sedentary older adults, underwent 24 one-hour exergame sessions over 12 weeks using the Nintendo Wii system. The study found that participants significantly improved their performance on the Wii tasks and showed greater improvements than a control group in measures of physical function, executive control, and processing speed. The researchers conclude that exergames, by combining cognitive and physical demands in an attractive activity, can effectively promote improvements in older adults.
- The study involved 30 independently living older adults (ages 65-78) who were previously sedentary and had never played video games.
- Participants showed a 97.5% adherence rate to the 12-week exergame training program, indicating high engagement.
- Exergame training resulted in significant benefits for executive control and processing speed tasks, with large effect sizes ($\eta^2 = 0.80$ and $\eta^2 = 0.79$ respectively), and also improved physical measures, except for one flexibility measure.

LO 3: Students will explore the applications of video games beyond entertainment, including their use in medicine, education, and social engagement.

EXPLORE FURTHER

Role of Video Games in Improving Health-Related Outcomes: A Systematic Review

RESEARCH REPORT | ~36 MIN

https://pmc.ncbi.nlm.nih.gov/articles/PMC3391574/

This systematic review, conducted in 2010, aimed to determine if video games are effective interventions for improving health outcomes. It analyzed 38 randomized controlled trials (RCTs) out of 1452 retrieved articles, focusing on studies where video games were the intervention and tested a health-promoting, clinically relevant outcome. The review found that video games were evaluated for a wide range of health purposes across all ages, most commonly for physical or psychological therapy. Among the 195 health outcomes examined across all studies, video games showed positive results for 69% of psychological therapy outcomes and 59% of physical therapy outcomes. However, the study noted that the overall quality of the studies was generally poor, with short follow-up periods and limited blinding or clear randomization methods. Despite these limitations, the review concludes that video games show potential for improving health across various areas and demographic groups.
- The review identified 38 Randomized Controlled Trials (RCTs) that tested the effect of video games on clinically relevant health outcomes.
- Video games were found to have improved 69% of psychological therapy outcomes and 59% of physical therapy outcomes across the included studies.
- Despite the promising results, the study highlighted that the quality of the research was generally poor, with two-thirds of studies having follow-up periods of less than 12 weeks and only 11% blinding researchers.

LO 3: Students will explore the applications of video games beyond entertainment, including their use in medicine, education, and social engagement.

EXPLORE FURTHER

Gamers Outreach

WEBSITE

https://gamersoutreach.org/

This is the web site for the organization that Zach Wigal founded.

LO 2: Students will be able to critically assess how video games reflect and shape cultural values, norms, and societal issues.
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EXPLORE FURTHER

Hermitcraft Gamers For Giving Charity Weekend! (Day 1)

YOUTUBE | 10 HRS 8 MIN

https://www.youtube.com/watch?v=788VKH4imec

Hermitcraft Gamers For Giving Charity Weekend! (Day 1) from impulseSV. Recording from an annual charity event where gamers raise funds for Gamers Outreach.

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EXPLORE FURTHER

Hermitcraft For Charity | Gamers Outreach Event Day 2!

YOUTUBE | 10 HRS 16 MIN

https://www.youtube.com/watch?v=UWsteNZKva8

Hermitcraft For Charity | Gamers Outreach Event Day 2! from PearlescentMoon. Recording from an annual charity event where gamers raise funds for Gamers Outreach.

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EXPLORE FURTHER

15 Ground-Breaking Video Games Modernising Healthcare

BLOG POST

https://www.chaostheorygames.com/blog/15-groundbreaking-video-games-modernising-healthcare

This article discusses how video games are transforming the healthcare industry, serving as progressive tools for patient care and professional development. It defines healthcare games as digital tools with a clinical purpose, designed to support medical professionals and aid patients, aiming to optimize care and health outcomes through gameplay. These games fall under the broader category of "serious games," which go beyond pure entertainment to change behavior, create emotional memories, or provide first-hand experiences of complex concepts. The article highlights three main reasons why healthcare games are effective: they allow for experimentation through simulation, help achieve real outcomes by enabling risk-free practice and measurable data, and modernize medicine by providing accessible and personalized treatment. It also categorizes health games into five types: Health and Wellness, Training and Simulation, Diagnosis, Treatment and Rehabilitation, and Education. It provides examples of 15 specific games from 2012 to 2022 that are changing lives in healthcare.
- Healthcare games have shown significant improvements in various therapy outcomes: 69% for psychological therapy, 59% for physical therapy, 50% for physical activity, 42% for health education, and 42% for pain distraction.
- Games can be developed to assist medical professionals or as stand-alone experiences for patients.
- Pokémon Go is cited as an outstanding example of a commercial entertainment game repurposed for health, with players collectively traveling 8.7 billion kilometers by walking in search of Pokémon as of December 2016.

LO 3: Students will explore the applications of video games beyond entertainment, including their use in medicine, education, and social engagement.

EXPLORE FURTHER

New Groundbreaking Study Finds Medical Video Games Improve Decision Making in Highly Experienced Doctors

ARTICLE

https://www.businesswire.com/news/home/20230131005473/en/%C2%A0New-Groundbreaking-Study-Finds-Medical-Video-Games-Improve-Decision-Making-in-Highly-Experienced-Doctors

This article reports on a groundbreaking study by Level Ex and CE Outcomes that found medical video games enhance the competence and clinical decision-making of highly experienced doctors. Unlike previous research that focused on medical students or residents, this study specifically targeted practicing physicians with an average age of 45 and 14 years of experience. The study demonstrated that doctors not only improved their in-game scores but also showed sustained improvement in clinical decision-making weeks after playing, indicating effective knowledge retention and transfer. Physicians in the study, who see an average of 151 patients a week, preferred learning through medical video games over traditional continuing medical education (CME). The specific game used, Top Derm, recreates medically accurate skin disorders, allowing doctors to practice diagnosis and treatment in a consequence-free setting.

- The study focused on highly experienced, busy practicing physicians (average 45 years old, 14 years in practice), a demographic challenging to reach for medical education.
- Playing Top Derm, a Level Ex game, for just 30 minutes led to a 12% increase in practical knowledge for doctors in some game modules.
- Nearly 75% of participating physicians preferred learning through medical video games over traditional continuing medical education (CME).

LO 3: Students will explore the applications of video games beyond entertainment, including their use in medicine, education, and social engagement.

EXPLORE FURTHER

Gaming and Medicine

ARTICLE

https://seasonedgaming.com/2022/09/07/gaming-and-medicine/

This article advocates for the benefits of video games in medicine, challenging common criticisms that view gaming as a waste of time. Written by a nurse, the piece highlights how games can keep patients occupied, especially pediatric patients, reducing boredom and restlessness in hospital settings. Beyond entertainment, the author explains that gaming is being used as therapy for various illnesses, citing EndeavorRX for ADHD in children and video game therapy for traumatic brain injury (TBI) to improve mobility and attention. The article also touches on gaming's positive impact on mental health by providing a sense of control for those feeling out of control in the real world. Additionally, it discusses the use of games for exercise (exergames) and for medical education, allowing professionals to practice complex procedures and decision-making in a risk-free virtual environment.

- EndeavorRX is an FDA-cleared video game designed to treat Attention Deficit Hyperactivity Disorder (ADHD) in children aged 8-12, serving as an adjunct to traditional treatments.
- Studies have shown that a 6-week video game therapy program improved balance, mobility, and selective attention in patients with traumatic brain injury (TBI).
- Companies like Level Ex create games for medical education and continuing medical education (CME) credits, allowing physicians to practice surgical maneuvers, diagnose patients, and make treatment choices in a simulated environment without risk to real patients.

LO 3: Students will explore the applications of video games beyond entertainment, including their use in medicine, education, and social engagement.

EXPLORE FURTHER

Serious Play Conference

WEBSITE

https://seriousplayconf.com/

A conference that explores “how the power of play, games and gamification are transforming education, training, and knowledge mobilization. Experience interactive sessions, compelling speakers, and cutting-edge research. This is where learning and innovation come to play.”

LO 3: Students will explore the applications of video games beyond entertainment, including their use in medicine, education, and social engagement.

EXPLORE FURTHER

Serious Games

ARTICLE

https://www.healthysimulation.com/serious-games/

This article explores the concept of serious games, defining them as computer-based programs that blend entertainment with learning through real-world scenario simulations. These applied games are highlighted for their significant utility in healthcare simulation education and training, improving skills, decision-making, and content retention for learners. The text further emphasizes how serious games enhance problem-solving, offer flexible remote learning options, and are increasingly used to train patients on health management. Various examples, such as SimPHARM, Vital Signs: ED, and Touch Surgery, illustrate the diverse applications of these game-based learning tools in medical contexts. Ultimately, it underscore the growing importance of serious games in transforming medical education and patient care, with events like the Serious Play Conference dedicated to their advancement.

LO 3: Students will explore the applications of video games beyond entertainment, including their use in medicine, education, and social engagement.

EXPLORE FURTHER

Journal of Medical Extended Reality

WEBSITE

https://home.liebertpub.com/publications/journal-of-medical-extended-reality/680

“Journal of Medical Extended Reality (J Med XR) is the first and preeminent peer-reviewed journal standing at the forefront of the burgeoning field of Medical Extended Reality (MXR), an interdisciplinary science focused on the role of spatial computing in healthcare. With the rapid advances in XR hardware and software, it is vital to examine how XR technologies, including virtual reality (VR), augmented reality (AR), and mixed reality (MR) can promote human health and wellbeing at scale.”

LO 3: Students will explore the applications of video games beyond entertainment, including their use in medicine, education, and social engagement.

EXPLORE FURTHER

XR and AI at Work: From Real - Time Guidance to Edge Deployment

YOUTUBE | ~50 MIN

https://www.youtube.com/watch?v=90xTrvTg-gQ

XR and generative AI are coming together to deliver real-time guidance in the flow of work. This Dell Technologies and NVIDIA panel explores how AI-powered XR can help users interpret data, understand physical environments, and access expert support hands-free, especially where speed, privacy, and local compute matter. Healthcare and life sciences serve as examples, including AI-enabled lab workflows from

Stanford that could accelerate life-saving therapies, and augmented reality surgical guidance from the University of Pittsburgh/UPMC designed to help surgeons work faster and smarter. The discussion also highlights how leading university labs are becoming launchpads for commercialization.

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